

Semester Operational Readiness Pack

Purpose: A production-readiness and company-operating template for Semester, a unified university operating system. Use this pack to determine whether a platform release, institutional pilot, integration, module, or company launch is ready for real users and customers.

Rule: A feature is not operational because it is deployed. It is operational only when it has a named owner, tested controls, observability, support path, recovery plan, evidence, and a signed go/no-go decision.

1. Master operating plan template

Create one master register. Every strategic initiative, module, integration, operational control, customer implementation, and launch requirement receives a row.

Field	Definition
Initiative ID	Stable identifier, e.g., INIT-OPS-001
Initiative	Short descriptive title
Workstream	Product & Engineering / Trust & Compliance / Customer Delivery / Commercial / Corporate Operations
Strategic outcome	Specific customer, business, or risk outcome
Scope	What is included and explicitly excluded
Customer/module/tenant impact	Who is affected and how
Accountable owner	Directly Responsible Individual (DRI)
Backup owner	Named secondary owner
Contributors	Teams, vendors, advisors, customer stakeholders
Priority	P0, P1, P2, P3
Risk tier	Low, Moderate, High, Critical
Dependencies	Systems, contracts, vendors, approvals, data, staffing
Budget	One-time and recurring cost
Target date	Planned completion date
Launch gate	Required gate or evidence before release
Evidence required	Tests, sign-offs, policies, screenshots, reports, runbooks
Success metric	Measurable completion/value indicator

Field	Definition
Status	Not started / Discovery / Planned / In progress / Blocked / Ready for review / Released / Validated
Risks/exceptions	Links to risk and exception records
Next review	Date and decision owner

Master-plan workstreams

Workstream	Mission	Example initiatives
Product & Engineering	Deliver a coherent, accessible, reliable platform	Shared object model, Today, LMS, Study Studio, grade ledger, integrations, performance
Trust & Compliance	Prove and maintain security, privacy, accessibility, AI, and interoperability controls	HECVAT, TrustEd Apps readiness, data inventory, AI governance, ACR/VPAT, evidence vault
Customer Delivery	Configure, launch, train, support, measure, renew, and offboard customers	Implementation factory, support service, success plans, pilot measurement, customer health
Commercial	Build predictable demand and revenue	Positioning, pricing, CRM, pilots, proposals, contracts, partnerships, renewals
Corporate Operations	Operate a durable company	Entity, IP, finance, insurance, people, vendors, board/advisor governance

2. Core workstream ownership

RACI-like leadership model

Workstream	Accountable owner	Must own	Required operating cadence
Product & Engineering	Chief Product/Technical lead	Roadmap, architecture, quality, delivery, performance, technical debt	Weekly delivery/reliability review
Trust & Compliance	Security/Privacy/Trust lead	Risk register, control evidence, access review, vendors, AI, accessibility, audit readiness	Weekly risk triage; monthly control review
Customer Delivery	Customer Success/Implementation lead	Onboarding, configuration, training, support, adoption, pilot value, renewals	Weekly customer health review
Commercial	Founder/Revenue lead	ICP, pipeline, pricing, proposals, contracts, partnerships, forecasting	Weekly pipeline/forecast review
Corporate Operations	Founder/Operations/Finance lead	Entity, banking, bookkeeping, insurance, hiring, contracting, vendor management, board records	Monthly close/operating review

Cross-functional decision owners

Decision domain	Accountable	Required reviewers
New product feature	Product lead	Engineering, design/research, accessibility, security/privacy, support
New AI model/provider/tool action	AI governance lead	Security, privacy/legal, engineering, accessibility, product
New integration/data flow	Integration lead	Security, privacy, product, customer delivery, customer IT owner
High-risk grading/assessment capability	Academic product owner	Accessibility, security/privacy, AI governance, customer academic owner
Sensitive support/basic-needs workflow	Student-support product owner	Privacy/legal, security, accessibility, institutional service owner
Production release	Engineering/release owner	Product, security, accessibility, operations, support
Customer go-live	Implementation lead	Customer sponsor, product, integrations, support, security/privacy as applicable
Commercial exception/discount	Revenue lead	Finance, legal, delivery, executive owner
Risk acceptance	Executive risk owner	Security/privacy/legal/accessibility/AI owner as applicable

Minimum DRI rules

- Every production system has one accountable owner and one backup owner.
- Every customer implementation has one Semester owner and one customer owner.
- Every integration has technical, data, and operational owners.
- Every data source has a source owner and freshness SLA.
- Every runbook has an owner, last-tested date, review date, and escalation path.
- No critical responsibility may be held only in one person's memory.

3. Seven pillars of production safety

Pillar 1 — Reliability and failure behavior

Question: Does the service remain understandable, safe, and recoverable when a dependency fails?

- [] Critical user journeys are documented.
- [] Service tiers and SLOs are defined.
- [] Dependencies are inventoried and classified by criticality.
- [] Timeouts, retries, circuit breakers, and graceful-degradation behavior are defined.
- [] Error messages give users a next action and support route.
- [] Maintenance windows and change-freeze periods are defined for critical academic components.
- [] Status page and customer-communication process are operational.

- [] No unsupported claim of 24/7 or emergency response is made.
- [] Failure modes have user-visible Source/Scope/Status states, e.g., stale, delayed

Required evidence: critical-journey map, dependency inventory, SLO document, error-state tests, status-page test, runbooks.

Pillar 2 — Data integrity and recovery

Question: Can Semester prevent, detect, correct, and recover from data loss, corruption, duplication, or stale integration data?

- [] Data classification and authoritative-source map are current.
- [] Every important object has source, freshness, version, owner, and audit metadata
- [] Database backups are encrypted, monitored, and restore-tested.
- [] RTO and RPO are defined for each service tier.
- [] Restore runbook is tested with recorded results.
- [] Migrations are versioned, reviewed, tested, and have rollback/repair plans.
- [] Critical writes are idempotent or protected by a write ledger.
- [] Grade, payment, consent, share, and integration writes are auditable and reconcil
- [] Deletion, retention, legal hold, export, and backup-expiry behavior is tested.
- [] Data reconciliation dashboard exists for material integrations.
- [] Dead-letter queues and manual repair workflows exist for failed messages/syncs.

Required evidence: backup configuration, restore record, migration logs, reconciliation report, deletion/hold test, grade/payment ledger tests.

Pillar 3 — Observability and incident response

Question: Can the team detect, understand, communicate, and resolve failures before they become prolonged customer harm?

- [] Centralized logs, metrics, traces, and audit events exist for production systems
- [] Monitoring covers availability, latency, errors, saturation, queue depth, databas
- [] Alerts have thresholds, owner, severity, runbook, and escalation path.
- [] Alert noise is reviewed and tuned.
- [] Incident severity levels, incident commander, communications owner, and customer
- [] Status page can be updated quickly by authorized staff.
- [] Incident runbooks exist for outages, data incidents, identity failure, integratio
- [] Tabletop exercises occur at least annually and after major operational changes.
- [] Post-incident reviews produce assigned, tracked corrective actions.

Required evidence: dashboard screenshots/configuration, alert test, incident plan, tabletop report, postmortem/corrective-action tracker.

Pillar 4 — Security, privacy, and access control

Question: Are users, tenants, data, keys, staff access, and production systems protected by enforceable controls?

- [] Production and non-production environments are separate.
- [] Secrets are stored in managed secret storage and never bundled to clients/source code.
- [] MFA protects privileged accounts.
- [] SSO, session controls, account recovery, access review, and offboarding are operational.
- [] Tenant isolation and role authorization have automated positive and negative tests.
- [] Private storage, signed URLs, upload validation, malware scanning, and content access control are implemented.
- [] Encryption in transit and at rest is verified.
- [] Vulnerability, dependency, secrets, and infrastructure scanning are enabled and operational.
- [] Production-access grants are least privilege, time-bound where possible, reason-based, and audited.
- [] Privacy data map, retention schedule, deletion/export workflow, and legal hold policies are implemented.
- [] Subprocessors and AI providers are inventoried and approved.

Required evidence: access review, IAM settings, authorization test suite, secrets scan, data map, vendor review, security scan/remediation log.

Pillar 5 — Performance, capacity, and cost resilience

Question: Can Semester meet user needs at academic peak times without unacceptable cost, latency, or failure?

- [] Performance budgets exist by critical page, API, device class, and network condition.
- [] Load tests cover start-of-term sign-in, assignment deadline, assessment autosave, and other high-risk scenarios.
- [] Database/query indexes, rate limits, queues, caching, CDN, and storage lifecycle management are implemented.
- [] Large lists/tables are virtualized or paginated.
- [] Search is indexed, debounced, permission-aware, and monitored.
- [] Media/file processing is queued with progress/failure/retry states.
- [] Low-bandwidth and lower-end-device testing is completed.
- [] Cost allocation exists by tenant, module, environment, AI use, storage, and integration.
- [] Budget alerts and cost-anomaly detection exist.
- [] Capacity and cost assumptions are reviewed before major customer launch.

Required evidence: load-test report, performance dashboard, cost report, device/network test, capacity plan, budget alerts.

Pillar 6 — Release, change, and dependency management

Question: Can Semester release safely, recover quickly, and prevent changes from breaking customers or compliance commitments?

- [] Protected branches, peer review, CI/CD checks, and deployment approval exist.
- [] CI includes type/unit/integration tests, authorization tests, secrets/dependency checks, and security scans.
- [] Staging mirrors relevant production behavior without ordinary production student data.
- [] Preview environments and synthetic test tenants exist.
- [] Every release has change scope, risk level, owner, rollback plan, customer impact assessment, and communication plan.
- [] Feature flags have owner, scope, expiry, rollout, and rollback details.
- [] Database migrations have validation and repair/rollback plans.

- [] Material changes trigger privacy, security, accessibility, AI, documentation, support, and other relevant changes
- [] Third-party dependency inventory includes health, contract, data, support, and other relevant information
- [] Certificate, domain, token, key, and license expiry are monitored.

Required evidence: CI configuration, release manifest, feature-flag register, dependency register, change approvals, rollback exercise.

Pillar 7 — User support, accessibility, and operational ownership

Question: Can a student, staff member, institution, or operator successfully use the service, get help, and understand what happens next?

- [] Critical workflows pass keyboard, screen-reader, zoom/reflow, mobile, contrast, and other accessibility requirements
- [] Accessibility feedback route, triage, remediation, workaround, and customer communication
- [] Help center, searchable knowledge base, in-product support, and customer-admin support
- [] Support cases have category, severity, owner, target response, escalation, and closure
- [] Student support is distinct from institutional technical support and security/privacy support
- [] Every user-facing domain displays source, scope, status, and authority where material
- [] Empty, loading, permission-denied, error, offline, stale-data, and recovery states
- [] Customer implementation, training, hypercare, adoption review, and renewal handoff
- [] Every module has product, operational, support, data/source, accessibility, privacy, and other relevant information

Required evidence: accessibility reports, support workflow, knowledge-base inventory, ticket SLA dashboard, training plan, ownership registry.

4. Unbuilt dependency register

A dependency is not “external” only. It includes unbuilt internal services, policies, people, content, integrations, legal approvals, vendors, customer decisions, and operational processes.

Field	Definition
Dependency ID	Stable identifier
Name	System, vendor, team, policy, data source, contract, or capability
Type	Internal service / external vendor / customer dependency / people / legal / content / infrastructure
Criticality	Critical / High / Moderate / Low
Owner	Semester and external/customer owner
Used by	Services, modules, customers, and workflows
Failure mode	What fails and how users are affected
Detection	Monitoring signal, health check, or manual check
Fallback	Graceful degradation/manual alternative
Recovery	Runbook and target recovery objective
Exit plan	Replacement, export, migration, or transition plan

Field	Definition
Contract/SLA	Terms, support, notice period, renewal/expiry
Last tested	Date
Next review	Date
Status	Healthy / At risk / Failed / Unknown

Required dependency categories

Cloud hosting and database
 Identity provider and SSO
 Email/SMS/push providers
 Payments processor
 AI model/provider and vector/retrieval components
 File storage and malware scanning
 CDN/DNS/domain/certificates
 Monitoring/logging/error tracking
 Support/ticketing/status-page provider
 Analytics/CRM/billing/accounting
 LMS/SIS/OneRoster/LTI/SCIM integrations
 Customer source-content owners and freshness commitments
 Security scanning and secrets management
 Legal/DPA/subprocessor/insurance dependencies
 Key-person and staffing dependencies

5. Recovery-runbook template

Every material incident scenario uses this template.

Runbook ID:
 Service/module/dependency:
 Severity trigger:
 Last tested:
 Owner / backup owner:

1. Detect
 - Alert, dashboard, customer report, or health check.
2. Triage
 - Scope, tenants affected, start time, user impact, data/security impact.
3. Contain
 - Feature flag, disable integration, rate limit, isolate tenant, revoke credential, freeze writes, switch to read-only mode.
4. Communicate
 - Internal channel, incident commander, executive/customer owner, status-page and customer-update templates.
5. Recover

- Exact technical steps, validation checks, reconciliation steps, rollback/restore procedure.

6. Validate

- User journey test, audit/reconciliation review, alert recovery, data-integrity check, customer confirmation where needed.

7. Close

- Incident record, root-cause analysis, corrective actions, owner/date, customer follow-up, evidence retention.

Minimum incident runbooks

Authentication/SSO failure
Database outage or corruption
Backup restore
Critical deployment rollback
Tenant-isolation/authorization concern
Data deletion or export failure
LMS/SIS/LTI/OneRoster sync failure
Grade/write reconciliation failure
Assessment autosave/submission failure
AI provider outage or unsafe-output escalation
Payment failure/refund/duplicate charge
Email/SMS/push delivery outage
File upload/malware scan failure
Security incident and suspected breach
DDoS/abuse/rate-limit event
Cloud region/provider outage
Critical vendor failure
Customer-admin misconfiguration

6. Launch readiness review gate

Decision outcomes

GO

All mandatory controls pass. Known limitations are low risk, documented, owned, and accurately communicated.

GO WITH CONDITIONS

No P0 risk remains. Time-bound P1/P2 conditions have owners, target dates, mitigation, customer communication, and executive acceptance.

NO-GO

A P0 or unresolved mandatory gate exists, including security/privacy breach risk, critical accessibility failure, data-loss/grade-integrity risk, unsupported claim, untested recovery path, or lack of accountable ownership.

Required launch review packet

Release/pilot/module scope and exclusions
Customer/tenant/cohort and contractual entitlement
Critical journeys and success metrics
Data flow, classification, source owner, retention, and integration map
Security/privacy/accessibility/AI risk assessments
Operational service tier, SLOs, RTO/RPO, monitoring, alerts, and on-call owners
Performance/capacity/load test results
Backup/restore, rollback, reconciliation, and failure-test evidence
Support, escalation, training, documentation, and hypercare plan
Known limitations, customer communications, and workaround plan
Open findings/exceptions with approval and expiry
Final approver names, date, and GO / GO WITH CONDITIONS / NO-GO record

Mandatory go/no-go checklist

Area	Gate	Required for GO
Scope	Enabled/excluded features are explicit	Yes
Ownership	Product, technical, operational, support, security/privacy/accessibility owners assigned	Yes
Data	Data map, permissions, retention, source/freshness, and export/deletion behavior documented	Yes
Security	Auth, tenant isolation, secrets, logging, vulnerability gates, incident route tested	Yes
Accessibility	Critical workflows tested; no unresolved critical barrier; alternatives documented	Yes
Reliability	SLOs, monitoring, alerting, status communications, dependency health, runbooks ready	Yes
Recovery	Backup restore, rollback, reconciliation, and failure behavior tested	Yes
Performance	Peak-load/capacity evidence meets defined thresholds	Yes for material workflows
AI	Provider, policy, data use, evaluation, monitoring, disable/rollback controls approved	If AI enabled
Integration	Sandbox validation, scopes, reconciliation, failure fallback, customer owner approved	If integration enabled
Support	Help, escalation, incident communication, implementation/hypercare plan ready	Yes
Commercial	Entitlement, price, invoicing, contract, support tier, renewal/exit terms ready	Yes
Evidence	All test results/sign-offs stored; open exceptions approved and time-bound	Yes

Final sign-off record

Launch ID:
Release/pilot/customer:
Decision: GO / GO WITH CONDITIONS / NO-GO
Decision date/time:
Decision chair:
Product sign-off:
Engineering sign-off:
Security sign-off:
Privacy/legal sign-off:
Accessibility sign-off:
AI governance sign-off, if applicable:
Operations/support sign-off:
Customer implementation sign-off:
Commercial/contract sign-off:
Conditions, if any:
Next review date:
Evidence package location:

7. Operating cadences

Cadence	Required review
Daily	Uptime, error/alert queue, security events, payments, support queue, integration health, critical customer issues
Weekly	Delivery/release review, risk triage, pipeline, customer health, support themes, vulnerability triage, cost anomalies
Monthly	Financial close, access review, evidence freshness, data-retention jobs, SLO/performance review, vendor review, roadmap review
Quarterly	Backup restore, incident tabletop, AI evaluation, accessibility review, risk-register review, customer value review, board/advisor update
Annually	Penetration test/independent security review, policy review, insurance renewal, full DR exercise, subprocessor review, accessibility documentation refresh

8. First operating-plan initiatives

ID	Initiative	Workstream	Priority	Mandatory evidence
INIT-OPS-001	Shared identity, tenant context, RBAC, audit events	Product & Engineering	P0	Authorization test suite, access review, audit samples
INIT-OPS-002	Data inventory, retention, deletion, legal-hold engine	Trust & Compliance	P0	Data map, deletion/hold/export tests
INIT-OPS-003	Observability, alerts, incident response, status communications	Product & Engineering	P0	Dashboards, alert test, incident tabletop

ID	Initiative	Workstream	Priority	Mandatory evidence
INIT-OPS-004	Backup/restore, DR, rollback, migration repair	Product & Engineering	P0	Restore exercise, rollback record, RTO/RPO
INIT-OPS-005	Accessibility design system and release gate	Trust & Compliance	P0	Manual/automated evidence, issue register
INIT-OPS-006	AI gateway, provider inventory, policy/evaluation controls	Trust & Compliance	P0	AI inventory, provider review, evaluations
INIT-OPS-007	Integration gateway, reconciliation, schema drift, contract tests	Product & Engineering	P0	Sandbox results, reconciliation dashboard, fallback tests
INIT-OPS-008	Support/implementation service and knowledge base	Customer Delivery	P1	SLA/workflow, training plan, support dashboard
INIT-OPS-009	Procurement room, contracts, evidence vault, HECVAT/TrustEd crosswalk	Trust & Compliance	P1	Current artifacts, completed assessment, evidence register
INIT-OPS-010	Pricing, CRM, billing, collections, forecasting	Commercial	P1	Price book, quote/order process, monthly forecast
INIT-OPS-011	Corporate entity, IP, bank, accounting, insurance, hiring controls	Corporate Operations	P0	Formation records, policies, insurance certificates
INIT-OPS-012	Pilot-to-platform implementation factory	Customer Delivery	P1	Readiness workbook, SOW, launch checklist, value review

9. Final standard

Build it.
 Test it.
 Secure it.
 Observe it.
 Recover it.
 Support it.
 Document it.
 Sell it honestly.
 Implement it repeatedly.
 Measure the outcome.

Semester becomes operational when these are not aspirations but repeatable, evidenced routines.